

■ Replace

`Replace[expr, rules]` applies a rule or list of rules in an attempt to transform the complete expression *expr*.

Example: `Replace[x^2, x^2 -> a] -> a`. ■ `Replace[x + 1, x -> a] -> 1 + x`. ■ The rules must be of the form *lhs -> rhs* or *lhs :> rhs*. ■ A list of rules can be given. The rules are tried in order. The result of the first one that applies is returned. If none of the rules apply, the original *expr* is returned. ■ If the rules are given in nested lists, `Replace` is effectively mapped onto the inner lists. Thus `Replace[expr, {{r11, r12}, {r21, ...}, ...}]` is equivalent to `{Replace[expr, {r11, r12}], Replace[expr, {r21, ...}], ...}`. ■ Delayed rules defined with `:>` can contain `/;` conditions. ■ `Replace[expr, rules]` applies rules only to the complete expression *expr*. You can use `Map` and `MapAt` to apply rules to specific parts of an expression, or you can use `ReplaceAll` to apply rules to all parts of an expression. ■ See page 207. ■ See also: `Rule`, `Set`, `AlgebraicRules`.